HCLTech Career Shaper Programme

III Year Industry Project — Technical Project Report

AgriScope — Agricultural Market Price Analyzer

Time-Series Forecasting of Wholesale Mandi Prices Using LightGBM with Micro-Climate Integration

Project Attribute	Specification / Official Record
Project ID & Track	P_085 / Track 05: Time-Series Market Analytics (Project No. 58)
Student / Learner	**Avnish Kumar** (Roll Number: 24003200100331)
Degree & Department	B.Tech in Computer Science & Engineering (3rd Year)
Institution / College	ABES Engineering College, Ghaziabad
Project Guide / Supervisor	Prof. / Dr. Poonam Jaswani, Department of CSE
Primary Price Dataset	AGMARKNET India Historical Mandi Prices (1,118,899 raw rows, 2024–2025)
Weather Integration	Open-Meteo Historical Archive API (13 physical market coordinates, 4,745 daily rows)
Evaluated Models	Seasonal Naive (Lag-7), LightGBM (Price-Only), LightGBM (Price + Weather), SARIMA
Evaluation Protocol	Strict chronological 56-day hold-out period (20 June 2025 to 14 August 2025)

Student & Supervisor Declarations

Student Declaration: I hereby declare that the project report entitled *'AgriScope — Agricultural Market Price Analyzer'* is an authentic record of original research and engineering carried out by me under the guidance of my supervisor. All data sources (AGMARKNET & Open-Meteo) and open-source libraries have been duly cited.
Student: Avnish Kumar (Roll No: 24003200100331) **Date:** 01-11-2026

Supervisor Certificate: This is to certify that this report has been prepared by Avnish Kumar under my supervision in satisfaction of the HCLTech Career Shaper Project requirements.
Project Guide: Prof. / Dr. Poonam Jaswani **Department:** Computer Science & Engineering

Executive Summary (Abstract)

Wholesale agricultural prices in Indian mandis suffer from severe volatility driven by harvest gluts, perishable crop spoilage, inter-regional transport friction, and monsoon weather shocks. AgriScope (Project ID: P_085) develops a leakage-free machine learning forecasting system combining 1,118,899 AGMARKNET daily mandi transaction records with Open-Meteo hyper-local historical weather data across 13 geographic locations. Following domain-specific price auditing (Min ≤ Modal ≤ Max) and daily median aggregation across 15 high-density mandi series (Wheat, Maize, Brinjal), 5,005 modeling observations were engineered with autoregressive lags (1, 7, 14, 30) and rolling statistics safeguarded by the shift(1) anti-leakage principle. Under a strict chronological 56-day hold-out test period (20 June to 14 August 2025), the deployed LightGBM (Price + Weather) model achieved MAE ■167.42, RMSE ■309.60, and MAPE 4.35% (~95.7% operational accuracy), representing a 38.4% error reduction over the Seasonal Naive baseline (MAE ■269.88, MAPE 6.41%). On the highly volatile Brinjal–Asansol series, LightGBM cut MAPE from 9.51% (SARIMA) to 4.20%. Weather revealed an empirical 'Weather Paradox': it reduced extreme outlier error spikes (RMSE improved by -2.43%) while slightly raising average error on calm days. An interactive Streamlit dashboard deploys recursive 1–30 day forecasts and spatial arbitrage diagnostics (disclosing a 48.8% price spread between Asansol and Birbhum).

Chapter 1: Introduction and Problem Formulation

1.1 Agricultural Price Volatility: Agricultural mandis operate as the primary wholesale clearing centers in rural India. Prices fluctuate drastically due to seasonal arrival gluts, high perishability of vegetables (spoiling in 2–3 days without cold storage), and logistical disruption during monsoon storms. Smallholder farmers frequently suffer distress selling due to information opacity.

1.2 Problem Statement (5W1H): Indian farmers and traders (Who) experience severe income losses from unpredictable price swings and regional market fragmentation (What) across wholesale mandis during seasonal transitions (When/Where) because price discovery is localized and opaque (Why). AgriScope develops a validated gradient-boosted time-series pipeline fusing mandi records with micro-climate weather to generate 1–30 day forecasts and arbitrage diagnostics via Streamlit (How).

1.3 Scope & Strategic Commodity Panel: Covers 3 distinct crops: (1) *Wheat* (MSP-backed durable staple); (2) *Maize* (feed/industrial coarse cereal); (3) *Brinjal* (highly perishable vegetable). Each commodity models the top 5 densest mandis (>80% annual date coverage), creating a robust 15-series panel.

Chapter 2: Literature Survey & Theoretical Background

2.1 Classical Econometric Time-Series (Box-Jenkins SARIMA): Assumes weak stationarity and linear autocorrelation. SARIMA models daily prices via seasonal autoregression (s=7 for weekly haat cycles). However, linear models catastrophically fail during abrupt non-linear supply shocks (e.g. monsoon flooding).

2.2 Tree-Based Gradient Boosting (LightGBM): Assembles additive regression trees using Gradient-based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB). LightGBM natively captures non-linear interactions (e.g., combining high rolling volatility with heavy rainfall) without requiring stationarity differencing.

2.3 Research Gaps Addressed: Prohibits random train-test shuffling (eliminating look-ahead bias), resolves target leakage in rolling statistics via shift(1), fuses market-level weather coordinates rather than national averages, and provides live multi-step recursive forecasting.

Chapter 3: System Architecture & Requirements

AgriScope is structured as a modular 3-tier architecture: (1) *Data Ingestion Layer:* AGMARKNET CSV ingestion, schema validation, domain auditing, and Open-Meteo REST API queries; (2) *Analytics & Modeling Layer:* Leakage-free lag/rolling calculation, chronological validation, and recursive multi-step forecasting; (3) *Presentation Layer:* Streamlit web application with native KPI cards, Plotly moving-average curves, and cross-mandi spatial comparison.

Chapter 4: Data Acquisition and Integrity Engineering

4.1 Profiling & Cleaning Protocol: The raw dataset contained 1,118,899 rows and 11 columns with zero nulls. The serial index (`Sl no.`) was removed. Dates were parsed as ISO format with `dayfirst=True` (zero corrupt dates).

4.2 Domain Bounds Validation (Min ≤ Modal ≤ Max):

- Dropped **470 rows** where minimum price exceeded maximum price (1,118,429 rows remaining).
- Dropped **730 rows** where modal price fell outside the [Min, Max] interval (1,117,699 clean rows remaining).

4.3 Aggregation & Short-Gap Regularization: Multi-grade daily trades were aggregated to single daily modal prices using the **Median** (robust against auction outlier skew). Series were regularized to continuous daily frequency (`asfreq('D')`). Internal gaps of ≤3 days were linearly interpolated (5,465 rows); long gaps were intentionally left unfilled to prevent artificial data synthesis.

Chapter 5: Feature Engineering & Anti-Leakage Safeguards

5.1 Feature Catalogue:

- *Autoregressive Lags:* `Lag_1` (momentum), `Lag_7` (weekly haat cycle), `Lag_14` (fortnightly trend), `Lag_30` (monthly anchor).
- *Rolling Statistics:* 7-day and 30-day rolling Mean (price level) and Standard Deviation (market turbulence).
- *Calendar Seasonality:* `Day_of_Week` (0–6), `Month` (1–12), `Week_of_Year` (1–52).
- *Micro-Climate Weather:* Mean/max/min temperature, precipitation sum, and max wind speed.

5.2 Mathematical Anti-Leakage Proof (The shift(1) Necessity): If today's price y_t is included in today's rolling 7-day mean, the feature becomes m_t = (1/7)*Σ(y_{t-i}). The learner easily inverts this to solve for the target: y_t = 7*m_t - Σ(y_{t-i}), achieving artificial 99% training accuracy that completely collapses in production. We enforce: **every rolling window is computed on shifted data: `shifted = group['Modal_Price'].shift(1)`**, guaranteeing ∂m/∂y_t = 0.

5.3 Weather Correlation vs Causation: Daily precipitation (+0.2327) and minimum temperature (+0.2067) correlate positively with price. In Indian agricultural logistics, rainfall floods rural roads, halts truck transport, and disrupts market arrivals, creating short-term supply scarcity rather than biological value changes.

Chapter 6 & 7: Experimental Modeling, Benchmark Results & Discussion

6.1 Chronological Evaluation: Enforces a strict 56-day hold-out period (20 June to 14 August 2025; Train: 4,166 rows, Test: 839 rows). Zero shuffling.

Model Architecture	Scope	MAE (■)	RMSE (■)	MAPE (%)	Performance Summary
Seasonal Naive Baseline (Lag-7)	Aggregate (15 Series)	■269.88	■538.92	6.41%	Simple weekly persistence heuristic
LightGBM (Price-Only)	Aggregate (15 Series)	■166.22	■317.32	4.27%	**38.4% error reduction** over baseline
LightGBM (Price + Weather) [DEPLOYED]	Aggregate (15 Series)	■167.42	■309.60	4.35%	**Weather Paradox:** -2.4% RMSE tail-risk
SARIMA (1,1,1)×(1,1,1,7)	Brinjal – Asansol	■440.26	■500.89	9.51%	Linear breakdown on vegetable shock

7.1 The Weather Paradox: Adding weather variables reduced RMSE (■317.32 → ■309.60, -2.43%), proving weather acts as insurance that trims extreme error spikes during stormy, road-flooded days. However, it added slight parameter noise on calm trading days, slightly raising MAE (■166.22 → ■167.42, +0.72%).

7.2 Feature Importance Decomposition: Recent price history (lags and rolling statistics) accounts for ~71% of total decision splits, micro-climate weather accounts for ~22%, and calendar seasonality accounts for ~7%. Short-term volatility (`Rolling_Std_7` with 1,055 splits) ranks second only to `Lag_1` (1,135 splits).

7.3 Root Cause Case Study (Brinjal in Asansol): Exhibits 3 real-world economic phases: (1) *Winter Glut (Jan–Feb):* Crash to ■1,200/quintal due to simultaneous harvest and zero cold storage; (2) *Heatwave (Apr–May):* Climb to ■3,500 due to 44.2

°C heatwave causing flower drop; (3) *Monsoon Shock (Jul):* Explosion to ■5,500 caused by 1,516 mm rain flooding fields and blocking delivery trucks.

7.4 Spatial Arbitrage Opportunity: On the latest common date, Brinjal cleared at ■4,300 in Asansol but ■6,400 in Birbhum—a **48.8% price spread (■2,100/quintal difference)** over 110 km, providing significant arbitrage potential for farmers using the dashboard.

Chapter 8: System Implementation & Dashboard Interface

The Streamlit application (`app.py`) provides a responsive decision-support platform served locally at `http://localhost:8501`. Three critical defects from initial prototypes were resolved: (1) Removed raw HTML string wrappers from `st.metric()` to restore native rendering; (2) Replaced `df.to_html()` tables with native `st.dataframe()`; (3) Replaced unstable `st.slider()` with `st.sidebar.number_input()` to eliminate cross-platform dynamic JavaScript chunk-loading crashes across cloud proxies.

Chapter 9 & 10: Conclusion, Future Scope & References

AgriScope demonstrates that leakage-free gradient-boosted trees outperform classical econometric models by 38.4% on agricultural time series. Future enhancements include extending history across multiple years, integrating Hindu festival arrival calendars via Prophet, and geocoding directly to mandi physical gates.

Key References:

[1] G. E. P. Box et al., *Time Series Analysis: Forecasting and Control*, 5th ed. Wiley, 2015.
[2] G. Ke et al., 'LightGBM: A Highly Efficient Gradient Boosting Decision Tree,' *NeurIPS 30*, 2017.
[3] AGMARKNET Portal, Directorate of Marketing & Inspection, Ministry of Agriculture, Govt. of India.
[4] Open-Meteo Global Meteorological Database, 'Historical Weather API,' open-meteo.com.